

An increasing number of researchers are exploring Influence Functions, a well-established concept from robust statistics, to better understand how training data influences the behaviour of machine learning models. By examining how small changes in the training data affect the parameters learned by a model, Influence Functions provide a useful perspective on both the robustness and fairness of these systems.

The paper work of Koh and Liang (2017) demonstrated that the Influence Function can be adapted to model machine learning models, including non-convex neural networks, to trace a prediction back to the training samples that influenced it the most. Their results show which specific training examples had the biggest impact on a prediction. If a model bases its decisions on strange, incorrect, or unrepresentative data, then its predictions cannot be considered fair or reliable. Koh and Liang also show that Influence Functions can spot mislabeled or harmful training samples. Moreover, the authors show that IF can detect mislabeled or harmful training points, providing a useful tool for detecting data quality problems that can hurt the model's performance.

Building on the framework, Deng et al. (2020) show how Influence-Function can be extended to analyse the behaviour of models trained with robust optimization. Instead of studying how a single training point affects the model, they derive the *Adversarial Influence Function* (AIF), which provides a formula that tells us how robust training shifts the model's parameters and how much the prediction loss on normal data increases as a result. This allows the authors to quantify *model sensitivity*, meaning how fragile or stable a model becomes after robust optimization. They use AIF to study how factors such as model complexity or randomized smoothing influence this sensitivity, and they show that the method works well even for kernel-based models like neural tangent kernels. They demonstrate that influence-function-based tools can be used to measure how training procedures affect the stability and reliability of model predictions.

Koh et al. (2019) investigate whether influence functions remain informative when applied to coherent groups of training points rather than isolated examples. Although the first-order approximation underlying IF is theoretically valid only for infinitesimal perturbations, the authors find that, in practice, the predicted influence of a group is strongly correlated with its actual effect, even as the absolute and relative errors grow large. This empirical observation holds across a variety of datasets and grouping strategies, with Spearman correlations of 0.89 to 0.99 in all settings. These results are practically significant because influence functions have been used to investigate model bias, audit the impact of demographic or otherwise grouped data sources, and assess vulnerability to data poisoning attacks. The ability to estimate the effect of removing entire segments of the training data without costly retraining is therefore directly relevant to understanding both robustness and fairness: a model whose predictions depend disproportionately on a small or unrepresentative subset of the data is more fragile under distribution shift and more likely to treat groups unequally.

Hurlin, Pérignon and Saurin (2024) study the fairness of credit scoring models, asking whether algorithms designed to distinguish good borrowers from bad ones also end up discriminating against individuals sharing a protected attribute such as gender, age or racial origin. They argue that such discrimination can be unintentional and can originate either from the training dataset itself or from the model structure. To address this, they develop a formal testing framework that allows practitioners to detect whether a scoring model violates fairness criteria, pinpoint the specific variables driving any detected unfairness, and then correct for it while preserving as much predictive accuracy as possible. They apply their framework empirically and show that it provides actionable guidance both for lenders seeking to self-audit their models and for regulators overseeing them. Their key finding that discriminatory outcomes can be traced back to the composition of the training data rather than to model design choices alone provides a direct motivation for data attribution methods such as influence functions. If harmful bias enters the model through specific training examples, then tools capable of identifying and quantifying the influence of those examples become essential for building scoring systems that are both accurate and fair.

Although these works are not specifically applied to credit-risk modelling, they show that influence-function-based methods provide a unified framework for analysing the stability, sensitivity, and reliability of predictive models. These properties are essential in credit scoring, where the justness of decisions, meaning their consistency, robustness to perturbations, and independence from biased data is a major regulatory concern. Influence-function approaches appear as promising tools for improving the quality of default-prediction models.

Moreover, the reliability of influence function estimates is not unconditional. Basu, Pope and Feizi (2021) conduct a large-scale empirical study of influence functions across a range of neural network architectures and datasets, including MNIST, CIFAR-10 and ImageNet. Their findings reveal that influence estimates are reasonably accurate for shallow networks but deteriorate significantly as network depth and width increase. These results have direct methodological implications for our study. To empirically validate the quality of our influence function estimates, we follow the Leave-One-Out retraining strategy from Koh and Liang (2017), the same approach adopted by Basu et al., measuring the extent to which IF scores approximate the true effect of removing individual training points on model predictions. We note, however, that Basu et al. observe that even LOO retraining can produce noisy ground-truth estimates at large scale (e.g., ImageNet), so we interpret the resulting correlations as a relative benchmark rather than an absolute gold standard.

Source

Basu, S., Pope, P., & Feizi, S. (2020). Influence functions in deep learning are fragile. *arXiv preprint arXiv:2006.14651*.

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